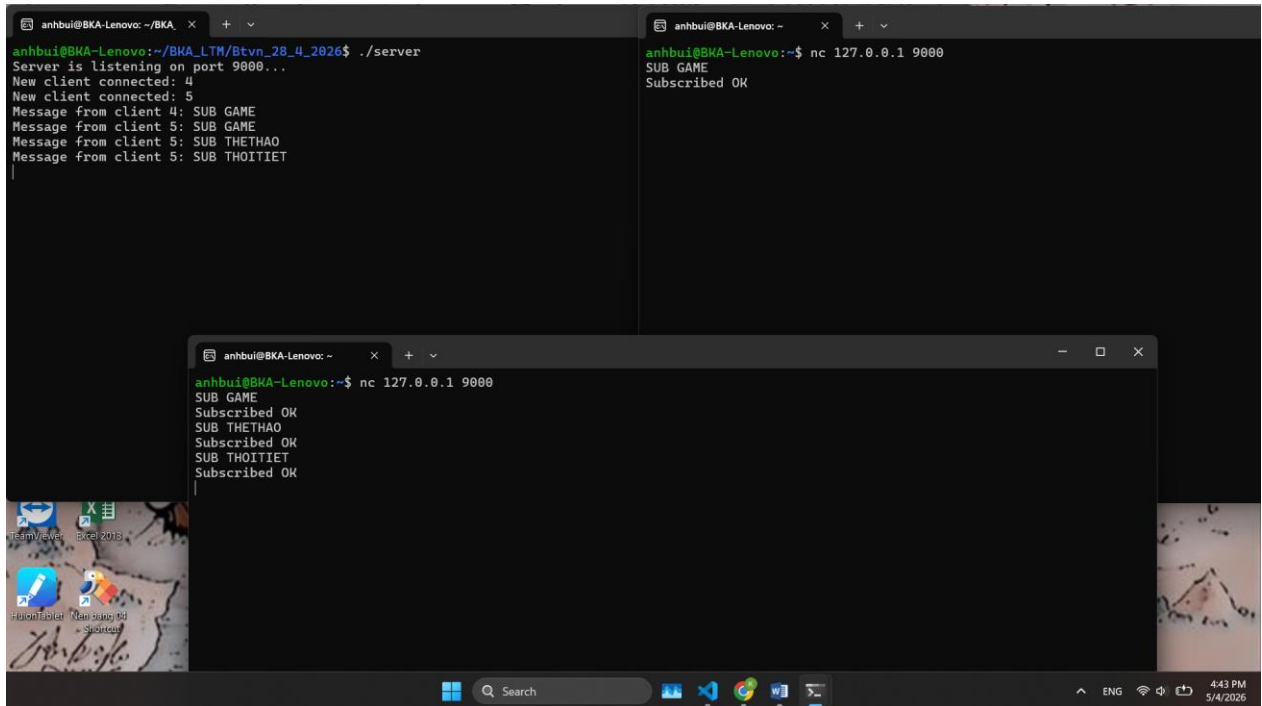


Báo cáo bài tập về nhà 28/04/2026

Hai client kết nối tới server và đăng kí topic, có thể đăng kí nhiều topic:

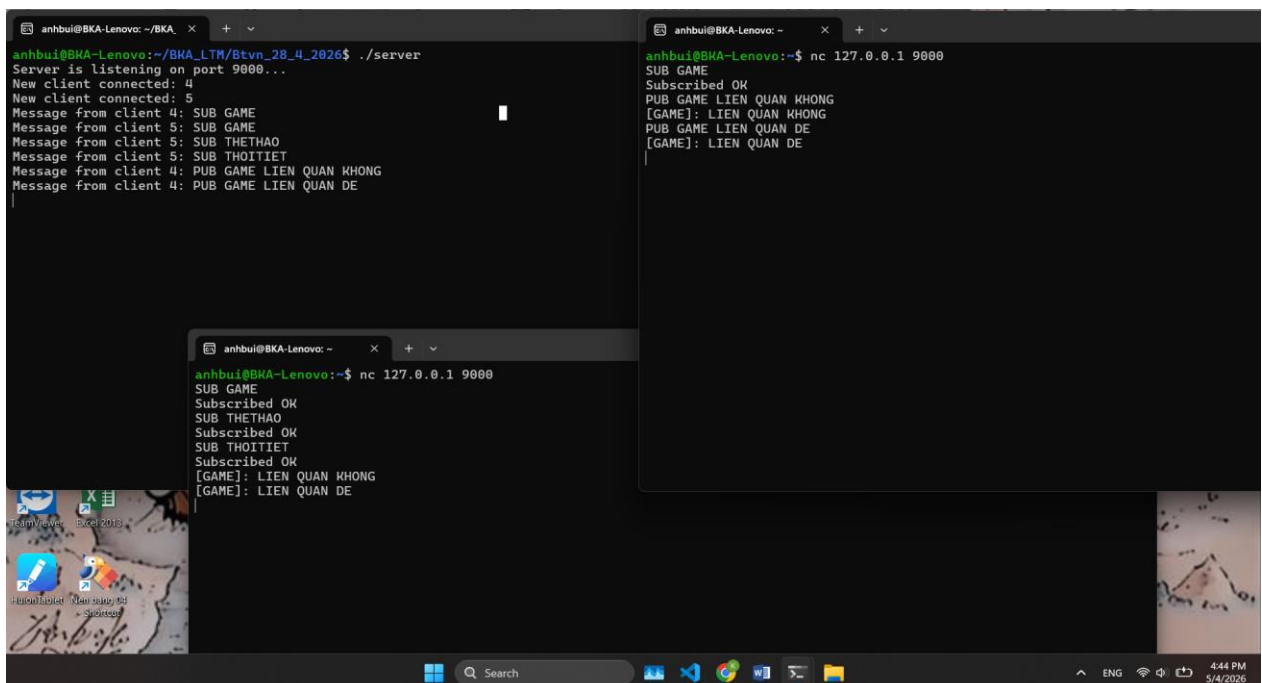


```
anhbui@BKA-Lenovo: ~/BKA
anhbui@BKA-Lenovo:~/BKA_LTM/Btvm_28_4_2026$ ./server
Server is listening on port 9000...
New client connected: 4
New client connected: 5
Message from client 4: SUB GAME
Message from client 5: SUB GAME
Message from client 5: SUB THETHAO
Message from client 5: SUB THOITIET

anhbui@BKA-Lenovo: ~
anhbui@BKA-Lenovo:~$ nc 127.0.0.1 9000
SUB GAME
SUB THETHAO
Subscribed OK

anhbui@BKA-Lenovo: ~
anhbui@BKA-Lenovo:~$ nc 127.0.0.1 9000
SUB GAME
Subscribed OK
SUB THETHAO
Subscribed OK
SUB THOITIET
Subscribed OK
```

Khi client gửi “PUB <topic> <msg>” gửi tin nhắn msg đến chủ đề topic, server nhận tin nhắn msg và chuyển tiếp cho tất cả các client đã đăng ký chủ đề topic trước đó

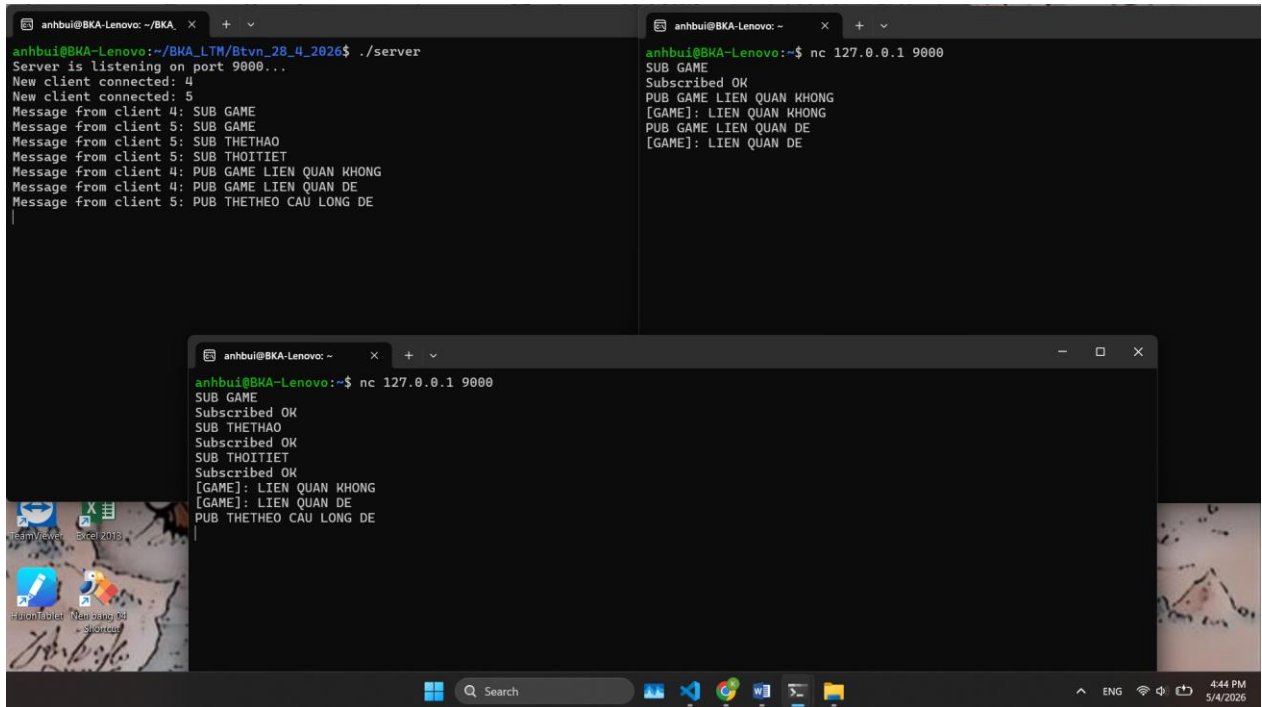


```
anhbui@BKA-Lenovo: ~/BKA
anhbui@BKA-Lenovo:~/BKA_LTM/Btvm_28_4_2026$ ./server
Server is listening on port 9000...
New client connected: 4
New client connected: 5
Message from client 4: SUB GAME
Message from client 5: SUB GAME
Message from client 5: SUB THETHAO
Message from client 5: SUB THOITIET
Message from client 4: PUB GAME LIEN QUAN KHONG
Message from client 4: PUB GAME LIEN QUAN DE

anhbui@BKA-Lenovo: ~
anhbui@BKA-Lenovo:~$ nc 127.0.0.1 9000
SUB GAME
Subscribed OK
SUB THETHAO
Subscribed OK
SUB THOITIET
Subscribed OK
[GAME]: LIEN QUAN KHONG
[GAME]: LIEN QUAN DE

anhbui@BKA-Lenovo: ~
anhbui@BKA-Lenovo:~$ nc 127.0.0.1 9000
SUB GAME
Subscribed OK
SUB THETHAO
Subscribed OK
SUB THOITIET
Subscribed OK
[GAME]: LIEN QUAN KHONG
[GAME]: LIEN QUAN DE
```

Server không chuyển tin nhắn tới các client mà không đăng kí topic như minh họa dưới:

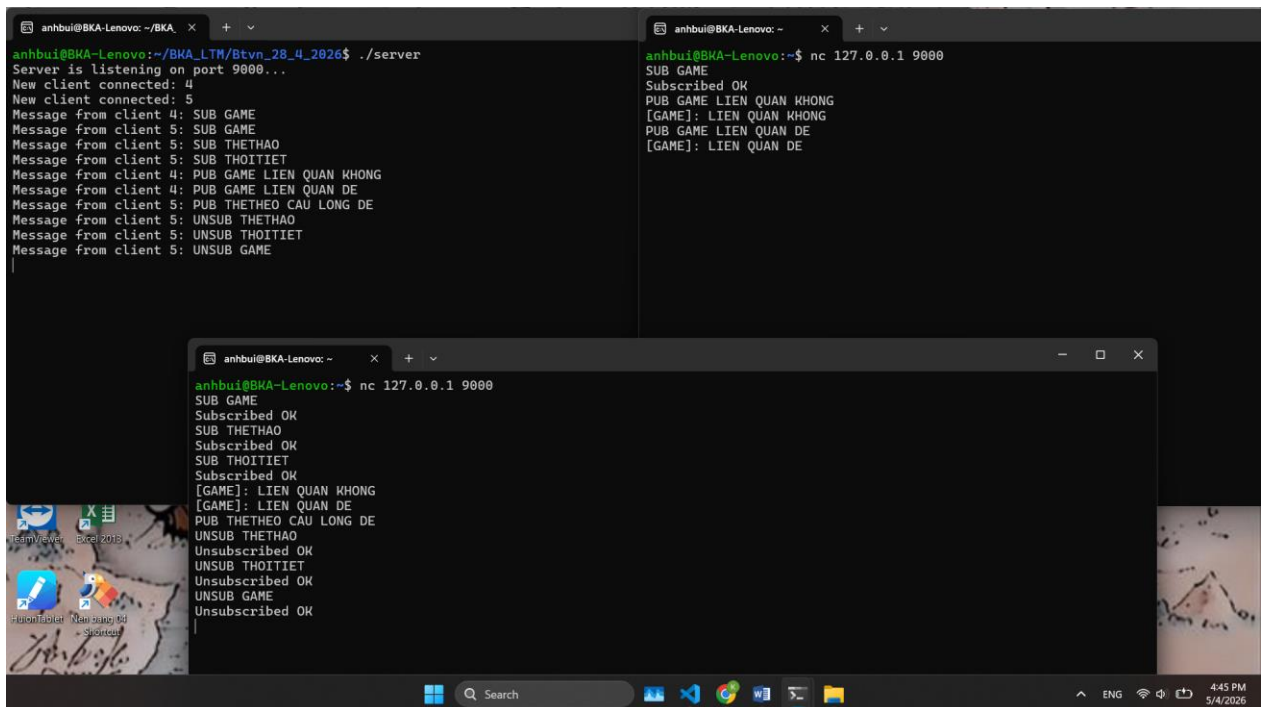


```
anhbui@BKA-Lenovo: ~/BKA
anhbui@BKA-Lenovo:~/BKA_LTM/Btvm_28_4_2026$ ./server
Server is listening on port 9000...
New client connected: 4
New client connected: 5
Message from client 4: SUB GAME
Message from client 5: SUB GAME
Message from client 5: SUB THETHAO
Message from client 5: SUB THOITIET
Message from client 4: PUB GAME LIEN QUAN KHONG
Message from client 4: PUB GAME LIEN QUAN DE
Message from client 5: PUB THETHEO CAU LONG DE

anhbui@BKA-Lenovo: ~$ nc 127.0.0.1 9000
SUB GAME
Subscribed OK
SUB THETHAO
Subscribed OK
SUB THOITIET
Subscribed OK
[GAME]: LIEN QUAN KHONG
[GAME]: LIEN QUAN DE
PUB THETHEO CAU LONG DE

anhbui@BKA-Lenovo: ~$ nc 127.0.0.1 9000
SUB GAME
Subscribed OK
SUB THETHAO
Subscribed OK
SUB THOITIET
Subscribed OK
[GAME]: LIEN QUAN KHONG
[GAME]: LIEN QUAN DE
PUB THETHEO CAU LONG DE
```

Client hủy đăng ký topic:

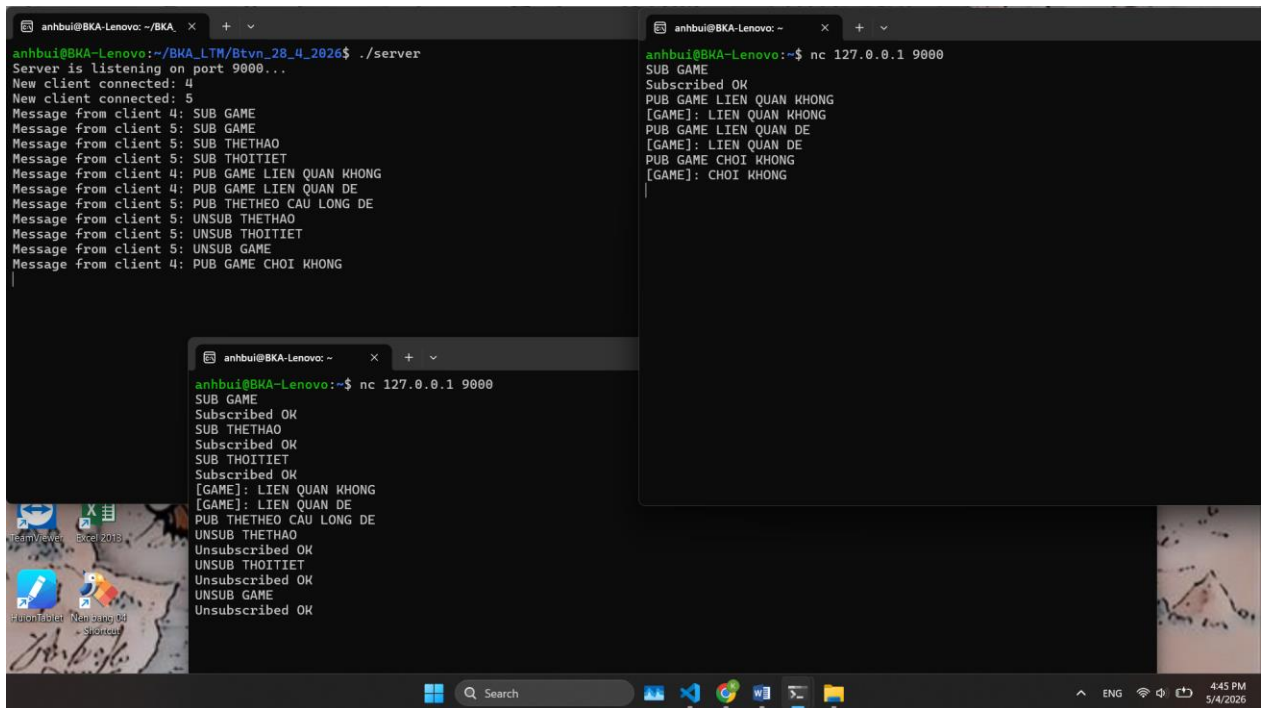


```
anhbui@BKA-Lenovo: ~/BKA
anhbui@BKA-Lenovo:~/BKA_LTM/Btvm_28_4_2026$ ./server
Server is listening on port 9000...
New client connected: 4
New client connected: 5
Message from client 4: SUB GAME
Message from client 5: SUB GAME
Message from client 5: SUB THETHAO
Message from client 5: SUB THOITIET
Message from client 4: PUB GAME LIEN QUAN KHONG
Message from client 4: PUB GAME LIEN QUAN DE
Message from client 5: PUB THETHEO CAU LONG DE
Message from client 5: UNSUB THETHAO
Message from client 5: UNSUB THOITIET
Message from client 5: UNSUB GAME

anhbui@BKA-Lenovo: ~$ nc 127.0.0.1 9000
SUB GAME
Subscribed OK
SUB THETHAO
Subscribed OK
SUB THOITIET
Subscribed OK
[GAME]: LIEN QUAN KHONG
[GAME]: LIEN QUAN DE
PUB THETHEO CAU LONG DE
UNSUB THETHAO
Unsubscribed OK
UNSUB THOITIET
Unsubscribed OK
UNSUB GAME
Unsubscribed OK

anhbui@BKA-Lenovo: ~$ nc 127.0.0.1 9000
SUB GAME
Subscribed OK
SUB THETHAO
Subscribed OK
SUB THOITIET
Subscribed OK
[GAME]: LIEN QUAN KHONG
[GAME]: LIEN QUAN DE
PUB THETHEO CAU LONG DE
```

Client sẽ không nhận được tin nhắn thuộc topic đã hủy:



```
anhbui@BKA-Lenovo: ~/BKA_LTM/Btvm_28_4_2026$ ./server
Server is listening on port 9000...
New client connected: 4
New client connected: 5
Message from client 4: SUB GAME
Message from client 5: SUB GAME
Message from client 5: SUB THETHAO
Message from client 5: SUB THOITIET
Message from client 4: PUB GAME LIEN QUAN KHONG
Message from client 4: PUB GAME LIEN QUAN DE
Message from client 5: PUB THETHEO CAU LONG DE
Message from client 5: UNSUB THETHAO
Message from client 5: UNSUB THOITIET
Message from client 5: UNSUB GAME
Message from client 4: PUB GAME CHOI KHONG

anhbui@BKA-Lenovo: ~$ nc 127.0.0.1 9000
SUB GAME
Subscribed OK
PUB GAME LIEN QUAN KHONG
[GAME]: LIEN QUAN KHONG
PUB GAME LIEN QUAN DE
[GAME]: LIEN QUAN DE
PUB GAME CHOI KHONG
[GAME]: CHOI KHONG

anhbui@BKA-Lenovo: ~$ nc 127.0.0.1 9000
SUB GAME
Subscribed OK
SUB THETHAO
Subscribed OK
SUB THOITIET
Subscribed OK
[GAME]: LIEN QUAN KHONG
[GAME]: LIEN QUAN DE
PUB THETHEO CAU LONG DE
UNSUB THETHAO
Unsubscribed OK
UNSUB THOITIET
Unsubscribed OK
UNSUB GAME
Unsubscribed OK
```